Integer Solutions of the Diophantine Equation $x^3 + y^3 = x^2 + kxy + y^2$

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Abstract

This paper is motivated by Problem 1 of the Balkan Mathematical Olympiad 2017, which asks for all positive integer solutions of

$$x^3 + y^3 = x^2 + 42xy + y^2.$$

We study the generalized family

$$x^3 + y^3 = x^2 + kxy + y^2, \qquad k \in \mathbb{Z},$$

over all integers, and denote by N_k the number of its integer solutions. We prove that for every pair of positive integers n and l there exists an integer $k > l$ such that $N_k > n$. On the other hand, for every $k \neq -1, 2$ the number of integer solutions is finite and satisfies the explicit bound

$$N_k \leq 4\,\tau\Big(\big|(k-2)^2(k+1)\big|\Big),$$

where $\tau(m)$ denotes the number of positive divisors of m; consequently $N_k \leq c\,|k|$ for an absolute constant c and all $|k| \geq 3$. For the exceptional values $k = -1$ and $k = 2$ the equation has infinitely many integer solutions. Finally, we show that there are infinitely many k for which the equation has exactly four integer solutions.

Introduction

The equation

$$x^3+y^3=x^2+kxy+y^2, \qquad k\in\mathbb{Z}, \tag{1}$$

with $k=42$ is Problem 1 of the Balkan Mathematical Olympiad (BMO) 2017 [2], where only *positive* integer solutions were required. In this paper we study all integer solutions of (1) for an arbitrary integer parameter k, and we write

$$N_k=\#\{(x,y)\in\mathbb{Z}^2:\ x^3+y^3=x^2+kxy+y^2\}.$$

For every k the pairs $(0,0)$, $(1,0)$ and $(0,1)$ satisfy (1); we call them the *trivial solutions*, so $N_k\geq 3$ for all k. For $k=42$ one finds $N_{42}=8$: besides the trivial solutions there are $(1,7)$, $(7,1)$, $(22,22)$ (the answer to the olympiad problem) and the two solutions $(-6,1)$, $(1,-6)$ with a negative coordinate.

Summary of results. For $k=-1$ and $k=2$ the equation has infinitely many integer solutions (see below). For every other k the number N_k is finite and is bounded by a divisor-function expression (Theorem 2), which yields a linear bound $N_k\leq c|k|$ (Theorem 4). On the other hand, N_k is unbounded as k varies (Theorem 1), while $N_k=4$ for infinitely many k (Theorem 3).

Small values of k

We first treat the values $k=0,-1,1,2$ by elementary means.

1. $k=0$**.** The equation is $x^3+y^3=x^2+y^2$. Put $Q=x^2-xy+y^2$, so that $x^3+y^3=(x+y)Q$. If $xy=0$, we get the solutions $(0,0)$, $(1,0)$, $(0,1)$. Let $xy\neq 0$. Since $Q\mid x^3+y^3$ we have $Q\mid x^2+y^2$, hence $Q\mid (x^2+y^2)-Q=xy$, and so $Q\leq|xy|$. But if $xy<0$ then $Q=x^2+y^2+|xy|>|xy|$, and if $xy>0$ then $Q-xy=(x-y)^2\geq 0$. Therefore $xy>0$ and $x=y$, and the equation

becomes $2x^3 = 2x^2$, i.e. $x = y = 1$.

Answer:

$$(x,y) = (0,0),\ (1,0),\ (0,1),\ (1,1).$$

2. $k = -1$**.** The equation is $x^3 + y^3 = x^2 - xy + y^2$, that is, $(x+y)Q = Q$ with Q as above, or $(x+y-1)Q = 0$. Hence either $x+y = 1$ or $Q = 0$, which over the integers means $x = y = 0$.

   Answer:

$$(x,y) = (0,0), \quad (x, 1-x)\ \ (x \in \mathbb{Z}).$$

3. $k = 1$**.** The equation is $x^3 + y^3 = x^2 + xy + y^2$. Let

$$s = x + y, \quad p = xy.$$

   Then

$$x^3 + y^3 = (x+y)^3 - 3xy(x+y) = s^3 - 3ps,$$

$$x^2 + xy + y^2 = (x+y)^2 - xy = s^2 - p.$$

   Substituting, we get $s^3 - 3ps = s^2 - p$, that is,

$$p(3s-1) = s^2(s-1).$$

   Since $\gcd(3s-1, s) = 1$ (a common divisor divides $(3s-1) - 3s = -1$), also $\gcd(3s-1, s^2) = 1$, and therefore $3s-1 \mid s-1$. Consequently

$$3s-1 \mid (3s-1) - 3(s-1) = 2,$$

   so $3s-1 \in \{-2,-1,1,2\}$. Only $3s-1 = -1$ and $3s-1 = 2$ give integer values of s, namely $s = 0$ and $s = 1$.

   Case 1: $s = 0$. Then $-p = 0$, so $p = 0$, and $x+y = 0$, $xy = 0$ give $x = y = 0$.

   Case 2: $s = 1$. Then $2p = 0$, so $p = 0$, and $x+y = 1$, $xy = 0$ give $(x,y) = (1,0)$ or $(0,1)$.

Answer:

$$(x,y) = (0,0),\ (1,0),\ (0,1).$$

4. $k = 2$**.** The equation becomes

$$x^3 + y^3 = x^2 + 2xy + y^2 = (x+y)^2.$$

Every pair with $x+y=0$ is a solution, so there are infinitely many integer solutions. If $x+y \neq 0$, dividing by $x+y$ gives $x^2 - xy + y^2 = x+y$, that is, $x^2 + y^2 = xy + x + y$. Doubling both sides, we obtain

$$(x-y)^2 + (x-1)^2 + (y-1)^2 = 2.$$

A sum of three squares equals 2 only if two of the squares are 1 and the third is 0. If $x = y$, then $(x-1)^2 = 1$, so $x = y = 2$ (the case $x = y = 0$ has $x+y=0$). If $x = 1$, then $(y-1)^2 = 1$, so $y = 0$ or $y = 2$. If $y = 1$, similarly $x = 0$ or $x = 2$.

Answer:

$$(x,y) = (x,-x)\ \ (x \in \mathbb{Z}), \quad (1,0),\ (0,1),\ (1,2),\ (2,1),\ (2,2).$$

Theorems and Proofs

For small $|k|$ the equation has only a few integer solutions (see the Introduction), yet Theorem 1 shows that the number of solutions is unbounded as k varies.

Theorem 1 (Almaty Math Battles 2026)**.** For all positive integers n and l there exists an integer $k > l$ such that the equation

$$x^3 + y^3 = x^2 + kxy + y^2$$

has more than n integer solutions.

Proof. We look for solutions of the form $y = mx$ with m a positive integer and $x \neq 0$. Substituting into

$$x^3 + y^3 = x^2 + kxy + y^2$$

gives

$$x^3 + (mx)^3 = x^2 + kx(mx) + (mx)^2,$$

hence

$$x^3(1 + m^3) = x^2(1 + km + m^2).$$

Therefore, if

$$1 + m^3 \mid 1 + km + m^2,$$

then

$$x = \frac{1 + km + m^2}{1 + m^3}, \qquad y = mx$$

is an integer solution.

So for each chosen m it is enough to require

$$1 + km + m^2 \equiv 0 \pmod{1 + m^3}.$$

A sufficient condition is

$$k \equiv m(m-1) \pmod{1 + m^3},$$

because then

$$1 + km + m^2 \equiv 1 + m^2(m-1) + m^2 = 1 + m^3 \equiv 0 \pmod{1 + m^3}.$$

Now choose

$$m_i = 2^{2^i} \qquad (i = 1, 2, \ldots, n+1),$$

and let

$$q_i = 1 + m_i^3 = 1 + 2^{3 \cdot 2^i}.$$

We claim that the integers $q_1, q_2, \ldots, q_{n+1}$ are pairwise coprime. Indeed, suppose $i < j$ and some integer $d > 1$ divides both q_i and q_j. Then

$$2^{3\cdot 2^i} \equiv -1 \pmod{d}.$$

Raising both sides to the power 2^{j-i} (an even number), we obtain

$$2^{3\cdot 2^j} \equiv (-1)^{2^{j-i}} = 1 \pmod{d}.$$

On the other hand, since $d \mid q_j$, we also have

$$2^{3\cdot 2^j} \equiv -1 \pmod{d}.$$

Thus $1 \equiv -1 \pmod{d}$, i.e. $d \mid 2$. But q_i is odd, so d is odd, and $d > 1$ is impossible. Hence

$$\gcd(q_i, q_j) = 1 \qquad \text{for all } i \neq j.$$

Therefore, by the Chinese Remainder Theorem, there exists an integer k such that

$$k \equiv m_i(m_i - 1) \pmod{q_i} \qquad (i = 1, 2, \ldots, n+1).$$

Moreover, all such integers form an arithmetic progression, so we may choose one with

$$k > l.$$

Since $l \geq 1$, we have $k > 0$.

For each i define

$$x_i = \frac{1 + km_i + m_i^2}{1 + m_i^3}, \qquad y_i = m_i x_i.$$

By construction, $x_i \in \mathbb{Z}$, hence $(x_i, y_i) \in \mathbb{Z}^2$. Moreover $x_i > 0$, because $k > 0$ makes the numerator positive. Also,

$$x_i^3 + y_i^3 = x_i^3(1 + m_i^3) = x_i^2(1 + km_i + m_i^2) = x_i^2 + kx_iy_i + y_i^2,$$

so (x_i, y_i) is a solution of the equation.

These solutions are distinct, since $x_i \neq 0$ and

$$\frac{y_i}{x_i} = m_i,$$

and the numbers $m_1, m_2, \ldots, m_{n+1}$ are pairwise distinct.

Thus the equation has at least $n+1$ distinct integer solutions. In particular, it has more than n integer solutions. □

Theorem 2. Let

$$N_k = \#\{(x,y) \in \mathbb{Z}^2 : \ x^3 + y^3 = x^2 + kxy + y^2\}.$$

If $k \neq -1, 2$, then

$$N_k \leq 4\,\tau\Big(\Big|(k-2)^2(k+1)\Big|\Big),$$

where $\tau(m)$ denotes the number of positive divisors of m.

Moreover, for $k = -1$ and $k = 2$ the equation has infinitely many integer solutions.

Proof. Set

$$s = x + y, \qquad p = xy.$$

Then

$$x^3 + y^3 = (x+y)^3 - 3xy(x+y) = s^3 - 3ps,$$

and

$$x^2 + kxy + y^2 = (x+y)^2 + (k-2)xy = s^2 + (k-2)p.$$

Hence

$$s^3 - 3ps = s^2 + (k-2)p,$$

so

$$s^3 - s^2 = p(3s + k - 2),$$

that is,

$$s^2(s-1) = p(3s + k - 2).$$

Let

$$D = 3s + k - 2.$$

Then

$$D \mid s^2(s-1).$$

Step 1: $D \neq 0$ for $k \neq -1, 2$. If $D = 0$, then $s^2(s-1) = 0$, so $s = 0$ or $s = 1$. Since $D = 3s + k - 2$, this gives $k = 2$ or $k = -1$, respectively. Therefore, for $k \neq -1, 2$ we have $D \neq 0$.

Step 2: $|D|$ divides $M := \left|(k-2)^2(k+1)\right|$. If $s = 0$, then $D = k - 2$ and the claim is clear. So let $s \neq 0$, and let $q^a \parallel D$ with q prime and $a \geq 1$. Since $D \mid s^2(s-1)$ and $\gcd(s, s-1) = 1$, the prime q divides exactly one of s and $s-1$.

Case 1: $q \mid s-1$. Since $q \nmid s$, from $q^a \mid s^2(s-1)$ we get

$$q^a \mid (s-1).$$

But also $q^a \mid D$, hence

$$q^a \mid D - 3(s-1) = k + 1.$$

Case 2: $q \mid s$. Write $q^b \parallel s$ with $b \geq 1$. Since $q \nmid s-1$, from $q^a \mid s^2(s-1)$ we obtain

$$a \leq 2b.$$

Also, because $q^a \mid D$ and $q^b \mid s$, we have

$$q^{\min(a,b)} \mid \gcd(D, s).$$

But

$$D - 3s = k - 2,$$

so

$$q^{\min(a,b)} \mid (k-2).$$

If $a \le b$, then $q^a \mid (k-2)$. If $a > b$, then $q^b \mid (k-2)$, and since $a \le 2b$, it follows that

$$q^a \mid q^{2b} \mid (k-2)^2.$$

Thus in Case 2 we always have $q^a \mid (k-2)^2$.

In both cases $q^a \mid M$. Since the prime powers $q^a \parallel D$ are pairwise coprime, we conclude that

$$|D| \mid M.$$

Step 3: counting. Since $k \neq -1, 2$, we have $M \neq 0$. Every integer solution (x,y) gives rise to an integer divisor D of M, and $s = x+y$ is then uniquely determined by

$$s = \frac{D-k+2}{3}.$$

After that, $p = xy$ is uniquely determined from

$$p = \frac{s^2(s-1)}{D}.$$

Finally, x and y are the roots of

$$t^2 - st + p = 0.$$

For fixed s and p this quadratic gives at most two ordered integer pairs (x,y), namely (x,y) and (y,x).

The number of integer divisors of M (positive and negative) is $2\tau(M)$. Hence

$$N_k \le 2 \cdot 2\tau(M) = 4\tau(M) = 4\,\tau\Big(\Big|(k-2)^2(k+1)\Big|\Big).$$

The exceptional values. As shown in the Introduction, for $k = -1$ every pair $(x, 1-x)$ with $x \in \mathbb{Z}$ is a solution, and for $k = 2$ every pair $(x, -x)$ with $x \in \mathbb{Z}$ is a solution. This completes the proof. $\square$

Lemma 1. Let p be a prime number. If p can be represented in the form

$$p = x^2 - xy + y^2$$

for some integers x, y, then either $p = 3$ or $p \equiv 1 \pmod 3$.

Proof. Modulo 3 we have $-xy \equiv 2xy$, hence

$$x^2 - xy + y^2 \equiv x^2 + 2xy + y^2 = (x+y)^2 \pmod 3.$$

Every square is congruent to 0 or 1 modulo 3, so $x^2 - xy + y^2 \equiv 0$ or $1 \pmod 3$ for all integers x, y, and never $2 \pmod 3$.

Now suppose that a prime p satisfies $p = x^2 - xy + y^2$. Then $p \equiv 0$ or $1 \pmod 3$. If $p \equiv 0 \pmod 3$, then $p = 3$ since p is prime. Otherwise $p \equiv 1 \pmod 3$. □

Theorem 3. There exist infinitely many integers k for which the equation

$$x^3 + y^3 = x^2 + kxy + y^2$$

has exactly four integer solutions (in particular, fewer than 5). More precisely, if $p \neq 2$ is a prime with $p \equiv 2 \pmod 3$ and $k = p - 1$, then the integer solutions are exactly

$$(0,0), \quad (1,0), \quad (0,1), \quad \left(\tfrac{p+1}{2}, \tfrac{p+1}{2}\right).$$

Proof. Let $p \neq 2$ be a prime with $p \equiv 2 \pmod 3$ and let $k = p - 1$. Note that p is odd, so $t = \frac{p+1}{2} = \frac{k+2}{2}$ is an integer. The pairs $(0,0)$, $(1,0)$, $(0,1)$ are solutions for every k, and (t,t) is a solution because $2t^3 = (k+2)t^2$. These four pairs are distinct.

Conversely, let $(x,y) \neq (0,0)$ be an integer solution and let $d = \gcd(x,y) > 0$. Write

$$x = dx_1, \quad y = dy_1, \quad \gcd(x_1, y_1) = 1.$$

Substituting into the equation and dividing by d^2, we get

$$d(x_1^3+y_1^3) = x_1^2+kx_1y_1+y_1^2.$$

Put $Q = x_1^2 - x_1y_1 + y_1^2 > 0$. Then $x_1^3+y_1^3 = (x_1+y_1)Q$ and $x_1^2+kx_1y_1+y_1^2 = Q+(k+1)x_1y_1$, so

$$d(x_1+y_1)Q = Q+(k+1)x_1y_1. \tag{2}$$

Hence $Q \mid (k+1)x_1y_1$. Now $\gcd(Q, x_1y_1) = 1$: a prime dividing x_1 and Q would divide $Q - x_1(x_1-y_1) = y_1^2$, hence y_1, contradicting $\gcd(x_1,y_1) = 1$; the case of a prime dividing y_1 and Q is symmetric. Therefore

$$Q \mid k+1 = p,$$

so $Q \in \{1, p\}$. By Lemma 1, the prime $p \equiv 2 \pmod 3$ is not of the form $x^2 - xy + y^2$, hence $Q \neq p$ and so $Q = 1$.

The equation $x_1^2 - x_1y_1 + y_1^2 = 1$ is equivalent to $(2x_1 - y_1)^2 + 3y_1^2 = 4$, so $y_1 \in \{-1, 0, 1\}$, and we find exactly six pairs:

$$(x_1,y_1) \in \{(1,0),(-1,0),(0,1),(0,-1),(1,1),(-1,-1)\}.$$

With $Q = 1$ and $k+1 = p$, equation (2) reads

$$d(x_1+y_1) = 1 + px_1y_1.$$

Recall $d > 0$.

- $(x_1,y_1) = (1,0)$: $d = 1$, giving $(x,y) = (1,0)$.
- $(x_1,y_1) = (0,1)$: $d = 1$, giving $(x,y) = (0,1)$.
- $(x_1,y_1) = (1,1)$: $2d = 1+p$, giving $(x,y) = \left(\frac{p+1}{2}, \frac{p+1}{2}\right)$.
- $(x_1,y_1) \in \{(-1,0),(0,-1),(-1,-1)\}$: the left-hand side is $-d$ or $-2d$ while the right-hand side is 1 or $1+p$, both positive; this is impossible for $d > 0$.

Together with $(0,0)$, these are exactly the four solutions listed.

Finally, there are infinitely many primes $p \equiv 2 \pmod 3$ (a special case of Dirichlet's theorem; an elementary argument: if $p_1, \dots, p_r$ were all of them, then $3p_1 \cdots p_r - 1 \equiv 2 \pmod 3$ is coprime to 3 and to every p_i, and not all of its prime factors can be $\equiv 1 \pmod 3$, a contradiction). Hence we obtain infinitely many values $k = p - 1$ with exactly four integer solutions. □

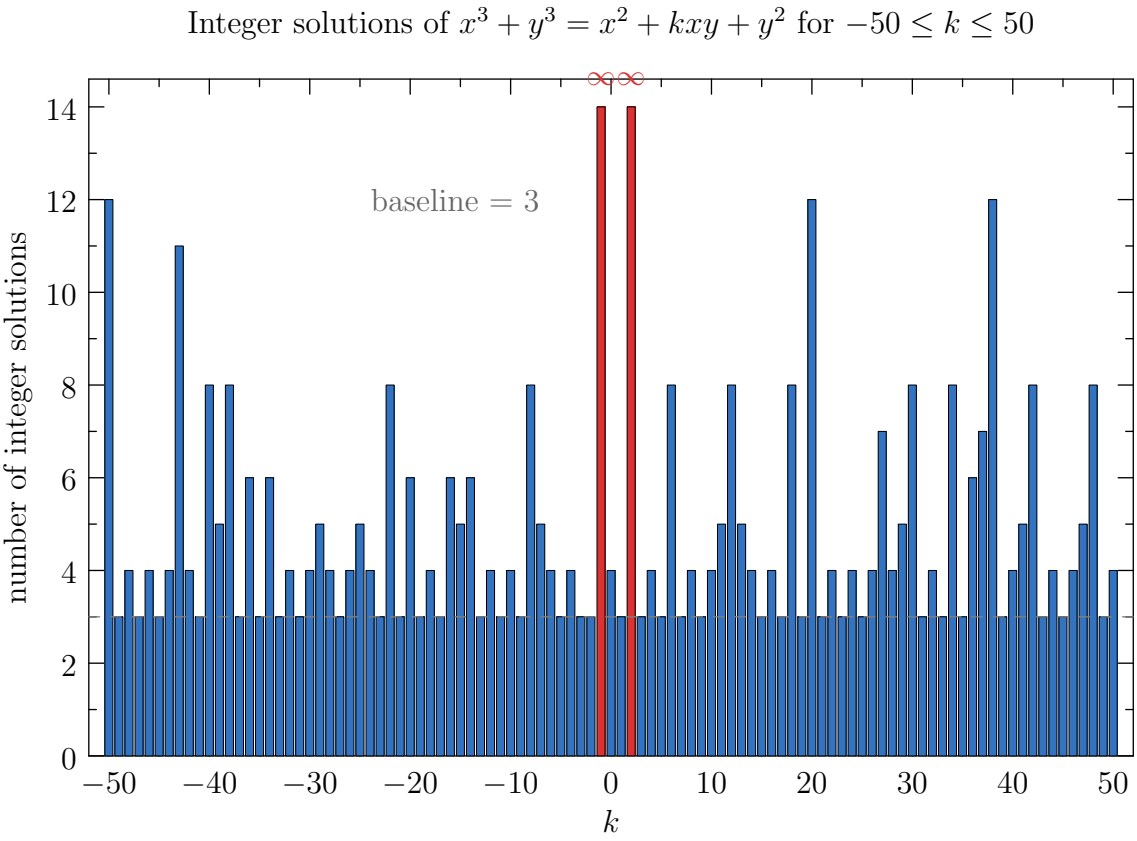


Figure 1: The number N_k of integer solutions for $-50 \le k \le 50$. The red bars mark $k = -1$ and $k = 2$, where $N_k = \infty$; the dashed line is the baseline $N_k \ge 3$ given by the trivial solutions.

Figure 1 shows N_k for $-50 \le k \le 50$. For $k \ne -1, 2$ in this range we have $3 \le N_k \le 12$, and $N_k \in \{3, 4\}$ for 66 of the 99 values of k. The trivial estimate $\tau(m) \le 2\sqrt{m}$ turns Theorem 2 into $N_k = O\left(|k|^{3/2}\right)$, which is much weaker than what the data suggest. A finer estimate of the divisor function gives a linear bound; the proof uses two small lemmas.

Lemma 2.1. For a prime p and an integer $a \geq 0$ put

$$f(p,a) = \frac{a+1}{p^{a/3}}.$$

If $p > 27$, then $f(p,a) \leq 1$ for all $a \geq 0$.

Proof. Since $p > 27$, we have $p^{1/3} > 3$, so $p^{a/3} \geq 3^a$. By Bernoulli's inequality, $3^a = (1+2)^a \geq 1+2a \geq a+1$. Hence $f(p,a) \leq \dfrac{a+1}{3^a} \leq 1$. □

Remark. The same argument works for every prime $p \geq 8$ (use $p^{1/3} \geq 2$ and $2^a \geq a+1$), so in fact only the primes $2,3,5,7$ can give $f(p,a) > 1$. The threshold 27 is kept for simplicity.

Lemma 2.2. For every prime p the function $a \mapsto f(p,a) = \dfrac{a+1}{p^{a/3}}$ is bounded on the non-negative integers. In particular, $C_p := \sup_{a \geq 0} f(p,a)$ is finite, and $C_p \geq f(p,0) = 1$.

Proof. For every $a \geq 0$,

$$\frac{f(p,a+1)}{f(p,a)} = \frac{a+2}{(a+1)\,p^{1/3}} = \frac{1+\frac{1}{a+1}}{p^{1/3}}.$$

Since $p^{1/3} > 1$, there is an a_0 such that $1 + \frac{1}{a_0+1} < p^{1/3}$, and then the ratio is less than 1 for all $a \geq a_0$. Hence $f(p,\cdot)$ is decreasing from a_0 onwards, and

$$\sup_{a \geq 0} f(p,a) = \max_{0 \leq a \leq a_0} f(p,a) < \infty.$$ □

Remark. For the primes $p < 27$ the maxima can be found directly: $C_2 = f(2,3) = 2$, $C_3 = f(3,2) \approx 1.442$, $C_5 = f(5,1) \approx 1.170$, $C_7 = f(7,1) \approx 1.046$, and $C_p = 1$ for $p = 11,13,17,19,23$. Figure 2 illustrates the case $p = 2$.

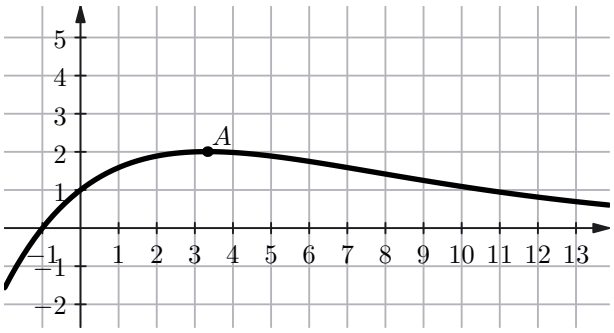


Figure 2: The graph of $x \mapsto (x+1)/2^{x/3}$, i.e. of $f(2,x)$; among the integers its maximum is $f(2,3) = 2$.

Theorem 4. There exists an integer constant c such that for every integer k with $|k| \geq 3$ the equation

$$x^3 + y^3 = x^2 + kxy + y^2$$

has at most $c\,|k|$ integer solutions, i.e. $N_k \leq c\,|k|$.

Proof. Let $|k| \geq 3$; in particular $k \neq -1, 2$. Put $M = \left|(k-2)^2(k+1)\right| \geq 1$ and write $M = \prod_i p_i^{a_i}$ with distinct primes p_i. Then $\tau(M) = \prod_i (a_i+1)$ and $M^{1/3} = \prod_i p_i^{a_i/3}$, so

$$\frac{\tau(M)}{M^{1/3}} = \prod_i f(p_i, a_i).$$

By Lemma 2.1 every factor with $p_i > 27$ is at most 1, and by Lemma 2.2 every factor with $p_i < 27$ is at most C_{p_i}. Since all $C_p \geq 1$, we get

$$\tau(M) \leq C M^{1/3}, \qquad C := \prod_{p<27} C_p.$$

For $|k| \geq 3$ we have $|k-2| \leq \frac{5}{3}|k|$ and $|k+1| \leq \frac{4}{3}|k|$, hence $M \leq \frac{100}{27}|k|^3$ and $M^{1/3} \leq \left(\frac{100}{27}\right)^{1/3} |k|$. By Theorem 2,

$$N_k \leq 4\tau(M) \leq 4C \left(\tfrac{100}{27}\right)^{1/3} |k|.$$

Thus the theorem holds with any integer $c \geq 4C \left(\frac{100}{27}\right)^{1/3}$. □

Remarks. (1) By the explicit values above, $C \approx 3.53$, so $c = 22$ works. (2) Replacing the exponent $\frac{1}{3}$ by an arbitrary $\varepsilon > 0$ in the definition of f, the same

argument gives $N_k = O_\varepsilon(|k|^\varepsilon)$ for every $\varepsilon > 0$. So Theorem 4 is far from sharp as an upper bound (for example we can use Wigert's works to obtain better upper bound); by Theorem 1, however, N_k is unbounded.

Conclusion

For equations of this form, it seems difficult to obtain a complete general description of all integer solutions. Nevertheless, the results proved above show that for every $k \neq -1, 2$, the equation has only finitely many integer solutions, while for the exceptional values $k = -1$ and $k = 2$, the set of integer solutions is infinite. Moreover, the number of solutions is unbounded as k varies (Theorem 1), but it grows at most linearly in $|k|$ (Theorem 4), and Theorem 3 shows that for infinitely many integers k the equation has exactly four integer solutions.

References

[1] Equations in China: Two Millennia of Innovation, Transmission and Re-transmission, Man-Keung.

[2] Balkan Mathematical Olympiad 2017, Problem 1 (Art of Problem Solving thread).

[3] On Prime Numbers of the Form a^2+ab+b^2 Where $a \in \mathbb{N}^+$, $b \in \mathbb{N}^+$, Arno Mikaelyan, Nairi Sedrakyan, Hayk Sedrakyan, Mathematical Reflections, 2025, no. 3.

[4] ON THE HYBRID MOMENTS OF THE GENERALIZED DIVISOR FUNCTION, Gergely Harcos